

ProActive Nine Week Implementation Plan

Executive summary

ProActive should be implemented as a **budget-conditioned active diagnostic-set learner**, not as a cheaper copy of HalluPrism. The professor’s planning document fixes the central framing: the system should treat multimodal reliability as **active diagnostic measurement**, adaptively acquire probe evidence under cost, and return a **calibrated diagnostic set** rather than a single confident source label. It also explicitly recommends a **full-evidence teacher**, a **sequential diagnostic state**, a **value-of-information policy**, a **GRU or small transformer encoder**, and a **conformal diagnostic-set head** with hard go/no-go criteria tied to cost–diagnostic frontiers and leave-model-out generalization.

HalluPrism is the right teacher substrate because it already defines the probe family and shows why a set-valued diagnostic output is needed: its visual, blank-image, grounding, and relation probes produce an entangled behavioral signature rather than cleanly separable causal labels, the full protocol costs roughly **7–8 forward passes per example**, and source-like coordinates overlap substantially under intervention. Those findings strongly support the professor’s “diagnostic sets with coverage” framing for ProActive.

The implementation plan below converts that framing into an executable nine-week program. The recommended version uses: a rule-backed **six-way behavioral target**; internal **multi-label evidence bits** retained as auxiliary supervision; **atomic probes** with cost measured primarily in **additional forward passes**; a **1-layer GRU** over clean and probe tokens; a **supervised counterfactual VOI learner** rather than RL; **split conformal APS** with **per-budget calibration** at 90% and 95% target coverage; core training data from **HallusionBench**, **POPE/RePOPE**, **VizWiz-VQA**, **VSR**, and a **GQA relation slice**; held-out shift tests on **PRE-HAL** and **IllusionBench**; and three open-weight MLLMs: **Qwen/Qwen3-VL-8B-Instruct**, **google/gemma-4-E4B-it**, and **OpenGVLab/InternVL3-9B**.

The main paper should aim to demonstrate four things. First, ProActive should approach the **full-evidence teacher** using materially fewer passes. Second, at matched 90% or 95% coverage, it should return **smaller calibrated diagnostic sets** than clean-only, random, and fixed-order baselines. Third, the learned acquisition policy should remain useful in **leave-one-model-out** evaluation. Fourth, it should show at least partial transfer on **held-out source-shift datasets** without overclaiming formal coverage under shift. That target matches the professor’s intended ICLR-level contribution and remains realistic within an 8–9 week execution window on A6000-class hardware.

Frozen scope and resolved design choices

What is fixed

The following design choices are already frozen by the project discussion and are preserved exactly in this plan:

Item	Final choice	Status
Project name	ProActive	Frozen
Primary target	Six-way behavioral diagnostic state: visual, language-prior, alignment/relation, mixed, unclear, no-failure	Frozen
Auxiliary target	Benchmark failure-family target	Frozen
Teacher	Full-evidence cache using all HalluPrism probes	Frozen
Actions	blank, blur, crop, brightness, noise, grounding, relation, STOP	Frozen
Primary cost	Additional forward passes	Frozen
Secondary cost	Normalized GPU latency	Frozen
Sequential encoder	GRU	Frozen
Policy learning	Supervised counterfactual VOI learning	Frozen
Budgets	$B \in \{1, 2, 3, 4, \text{full}\}$	Frozen
Conformal method	Split conformal APS per budget at 90% and 95%	Frozen
Split	Grouped 70/10/10/10 train/val/cal/test	Frozen
Core data	POPE/RePOPE, HallusionBench, VizWiz-VQA, VSR, GQA relation slice	Frozen
Held-out data	PRE-HAL, IllusionBench	Frozen
Models	Qwen3-VL-8B-Instruct, Gemma-4-E4B-it, InternVL3-9B	Frozen

These choices align directly with the professor’s ADS framing and with HalluPrism’s role as a full-probe source of behavioral evidence rather than a final deployment policy.

Resolved technical decisions

The most important technical ambiguity was the meaning of the target label. The strongest executable choice is:

Primary conformal target: a six-way **behavioral state** $f(x) \in \mathcal{F}$, where

$$\mathcal{F} = \{\text{visual, language-prior, alignment, mixed, unclear, no-failure}\}.$$

Internal auxiliary target: a three-bit behavioral evidence vector

$$b(x) = (b_V, b_L, b_A) \in \{0, 1\}^3,$$

which is used for auxiliary supervision and audit analysis but is **not** the final public prediction object.

This resolves two problems simultaneously. It preserves the professor’s set-valued output objective, and it respects HalluPrism’s central observation that visual fragility, blank-image persistence, and alignment instability are informative but entangled behavioral signatures, not independent causal truths.

The internal **bit construction** is recommended as follows:

$$\begin{aligned}
 b_V &= \mathbf{1} \left[\frac{1}{4} \sum_{k=1}^4 \text{flip}_k \geq 0.25 \quad \vee \quad \frac{1}{4} \sum_{k=1}^4 \Delta c_k \geq 0.15 \right] \\
 b_L &= \mathbf{1} \left[\text{blank_match} = 1 \quad \wedge \quad \frac{c_{\text{blank}}}{c_{\text{clean}} + \varepsilon} \geq 0.8 \right] \\
 b_A &= \mathbf{1} \left[\text{ground_flip} = 1 \quad \vee \quad \Delta c_{\text{ground}} \geq 0.15 \quad \vee \quad (\text{relation_available} \wedge \text{swap_invariance} = 1) \right].
 \end{aligned}$$

Then the six-way class is assigned by rule:

- **no-failure** if the clean answer is correct and $b_V + b_L + b_A = 0$
- **mixed** if $b_V + b_L + b_A \geq 2$
- **visual, language-prior, or alignment** if exactly one corresponding bit is active
- **unclear** otherwise

This is intentionally conservative. It avoids inventing a “clean source label” when the probe evidence is weak or contradictory. It also retains the bit-vector internally so the paper can report both multiclass set prediction and multi-label evidence reconstruction. That matches the professor’s coverage-oriented framing while remaining easy to audit. `filecite{turn0file2}{turn0file0}`

Model and dataset scope

The selected model IDs are current and technically reasonable for the plan. The Qwen model card exposes `Qwen/Qwen3-VL-8B-Instruct` under Apache 2.0, recommends the latest Hugging Face Transformers source build, and shows native multimodal loading through `Qwen3VLForConditionalGeneration`, with a recommendation to enable `flash_attention_2` for memory and speed. The Gemma card lists `google/gemma-4-E4B-it` under Apache 2.0, describes Gemma 4 as multimodal, and notes that the smaller models support image inputs and also audio/video handling in the family documentation. The InternVL card lists `OpenGVLab/InternVL3-9B`, identifies its language backbone as `internlm3-8b-instruct`, requires `transformers>=4.37.2`, and loads through `AutoModel` with `trust_remote_code=True`; the page also states the project is released under MIT.

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The core dataset suite is also coherent for the target problem. **POPE** provides object-presence hallucination stress; **RePOPE** is the corrected-label robustness companion. **HallusionBench** gives mixed visual illusion and entangled hallucination cases. **VizWiz-VQA** contributes real-user poor-image and answerability stress. **VSR** provides explicit spatial relations across more than 10k text–image pairs and 66 relation types, though the usable dev subset is smaller for quick experimental cycles. **GQA** is the right source for a synthetic relation slice because its scene graphs and functional programs make relation-controlled statement generation tractable. **IllusionBench** and **PRE-HAL** are well-suited held-out stress datasets because they focus on visual illusions and on perception–reasoning hallucination, respectively.

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The recommended **full-run sample caps** are:

Dataset role	Dataset	Per-model count	Reason
Core	HallusionBench	951	Use all image-paired examples
Core	POPE/RePOPE	3,000	1k each from random, popular, adversarial
Core	VizWiz-VQA	3,000	Enough answerability/quality diversity without full 4,319 cost
Core	VSR	340	Use full dev subset
Core	GQA relation slice	1,000	Adds controlled relation diversity
Held-out	PRE-HAL	600	Shift evaluation, not training
Held-out	IllusionBench	600	Shift evaluation, not training

That gives **8,291 core instances per model** and **9,491 total instances per model** including held-out stress tests. Across three models, the plan targets **24,873 core examples** and **28,473 total examples**. This cap is large enough for meaningful policy learning while remaining feasible in nine weeks on A6000 GPUs.

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Formal problem statement and algorithms

Problem formulation

For a model M , image I , prompt q , and clean generated answer $\hat{y}$, define the ProActive instance as

$$x = (I, q, \hat{y}, M).$$

The action space is

$$\mathcal{A} = \{\text{blank}, \text{blur}, \text{crop}, \text{brightness}, \text{noise}, \text{grounding}, \text{relation}, \text{STOP}\}.$$

Each non-stop action a yields a probe observation

$$o_a(x) = (\tilde{y}_a, \text{flip}_a, \Delta c_a, m_a^{\text{exact}}, m_a^{\text{sem}}, \text{app}_a, \tau_a),$$

where $\tilde{y}_a$ is the probe answer, flip_a indicates whether the normalized answer changed, Δc_a is the confidence shift, m_a^{exact} and m_a^{sem} are exact and semantic match indicators, app_a is the applicability mask, and τ_a stores probe metadata such as perturbation severity. HalluPrism already uses this style of perturbation–response object and explicitly measures answer flips and confidence shifts under visual, blank-image,

and alignment probes. [filecite0turn0file0](#)

ProActive begins with a cheap clean feature vector

$$g(x) = [c, \bar{H}, \bar{m}, \ell, \text{IQA}, \text{qtype}, \text{answerability_hint}, \text{relation_available}],$$

where c is the normalized answer probability, $\bar{H}$ is mean token entropy over the generated answer, $\bar{m}$ is mean token margin, ℓ is answer length, IQA is image-quality metadata, qtype is question-type metadata, and the relation flag indicates whether relation swapping is legal. HalluPrism already uses normalized answer probability and dataset-specific normalization as part of its clean measurement, and the professor’s document explicitly allows clean-pass features such as scalar confidence, answer length, entropy or margin, question type, and image-quality measures. [filecite0turn0file0](#)[turn0file2](#)

The sequential diagnostic state is

$$h_t = \text{GRU} (e_{\text{clean}}, e_{a_1, o_1}, \dots, e_{a_t, o_t}),$$

with remaining budget

$$B_t \in \{0, 1, 2, 3, 4, \text{full}\}.$$

The diagnostic head predicts

$$p_\theta(f \mid h_t) \in \Delta^6,$$

and the auxiliary head predicts

$$r_\theta(b \mid h_t) \in [0, 1]^3.$$

The calibrated set head returns

$$\Gamma_{\alpha, B}(h_t) \subseteq \mathcal{F},$$

with target marginal coverage

$$\mathbb{P}(f(x) \in \Gamma_{\alpha, B}(h_T)) \geq 1 - \alpha.$$

This is the exact mathematical object the professor’s plan calls for: a budget-conditioned, cost-aware diagnostic set with held-out calibration rather than a single forced source assignment. [filecite0turn0file2](#) [cite0turn12academia0](#)[turn12academia1](#)

Diagnostic loss

The recommended diagnostic training objective is

$$\mathcal{L}_{\text{diag}} = \underbrace{\text{CE}(f, p_\theta(\cdot \mid h_t))}_{\text{six-way class}} + \beta_{\text{bit}} \underbrace{\text{BCE}(b, r_\theta(\cdot \mid h_t))}_{\text{multi-label evidence}} + \beta_{\text{sig}} \underbrace{\|\hat{s}_\theta(h_t) - s_{\text{teacher}}(x)\|_2^2}_{\text{continuous signature reconstruction}},$$

where $s_{\text{teacher}}(x) = (V, L, A)$ is the full-evidence HalluPrism signature and $\hat{s}_\theta$ is an auxiliary regression head. The recommended default weights are

$$\beta_{\text{bit}} = 0.5, \quad \beta_{\text{sig}} = 0.1.$$

This choice keeps the primary task aligned with the paper’s six-way calibrated set objective, while auxiliary heads preserve the richer teacher evidence and improve representation quality under entanglement. That is consistent with the professor’s recommendation to preserve, not suppress, entangled structure. [filecite0turn0file2](#)[turn0file0](#)

Counterfactual value of information

The key implementation choice is to train VOI from **realized counterfactuals** using cached teacher outcomes, not from online RL. Because the teacher cache stores every probe result for every example, the effect of any candidate next action is known offline.

For a partially observed state h_t and a legal next action a , define the realized value target as

$$y_{\text{VOI}}(x, h_t, a) = \underbrace{\left(|\Gamma_{\alpha, B_t}^{\text{val}}(h_t)| - |\Gamma_{\alpha, B_t}^{\text{val}}(h_t^a)| \right)}_{\text{set-size gain}} + \eta_{\text{loss}} \underbrace{\left(\mathcal{L}_{\text{diag}}(h_t) - \mathcal{L}_{\text{diag}}(h_t^a) \right)}_{\text{diagnostic-loss gain}} - \lambda_{B_t} c(a),$$

where h_t^a is the state after applying the teacher observation for action a , Γ^{val} uses **validation-temporary** calibration rather than the final held-out calibration split, $c(a) \in \{0, 1\}$ for atomic probes, and $\eta_{\text{loss}} = 0.25$ is the recommended default. The primary term is calibrated-set shrinkage because the final output is a diagnostic set; the loss drop term stabilizes learning when many actions leave set size unchanged.

The professor’s document explicitly proposes VOI in terms of reduction in diagnostic loss or diagnostic-set size minus cost, and this combined form is the cleanest executable version of that idea. [filecite turn 0 file 2](#)

The policy network predicts $\hat{y}_{\text{VOI}}(h_t, a)$, and uses

$$a_t^* = \arg \max_{a \in \mathcal{A}_t} \hat{y}_{\text{VOI}}(h_t, a), \quad \mathcal{A}_t = \{a \in \mathcal{A} : \text{app}_a = 1, c(a) \leq B_t\}.$$

The stop rule is

$$a_t^* = \text{STOP} \quad \text{if} \quad \max_{a \in \mathcal{A}_t \setminus \{\text{STOP}\}} \hat{y}_{\text{VOI}}(h_t, a) \leq 0.$$

That converts the intuitive “run the next useful test if its value exceeds its cost” idea into a deterministic implementation rule. [filecite turn 0 file 2](#)

Policy loss

The recommended policy loss is a combined **ranking + regression + best-action classification** objective:

$$\mathcal{L}_{\text{policy}} = \mathcal{L}_{\text{rank}} + 0.25 \mathcal{L}_{\text{mse}} + 0.5 \mathcal{L}_{\text{act}}.$$

The terms are:

$$\mathcal{L}_{\text{mse}} = \frac{1}{|\mathcal{A}_t|} \sum_{a \in \mathcal{A}_t} (\hat{y}_{\text{VOI}}(h_t, a) - y_{\text{VOI}}(x, h_t, a))^2,$$

$$\mathcal{L}_{\text{act}} = \text{CE}(\arg \max_a y_{\text{VOI}}(x, h_t, a), \arg \max_a \hat{y}_{\text{VOI}}(h_t, a)),$$

and

$$\mathcal{L}_{\text{rank}} = \sum_{a \neq b} \max(0, m - \text{sgn}(y_a - y_b)(\hat{y}_a - \hat{y}_b)), \quad m = 0.05.$$

This is more robust than pure regression because policy quality depends mostly on **action ordering**, not on perfectly calibrated numeric VOI values.

APS and per-budget calibration

For a frozen diagnostic classifier and a fixed budget B , sort class probabilities in descending order:

$$\pi_{(1)}(h) \geq \pi_{(2)}(h) \geq \dots \geq \pi_{(6)}(h).$$

If the true label rank on calibration instance i is ρ_i , define the APS score

$$s_i^{(B)} = \sum_{j=1}^{\rho_i} \pi_{(j)}(h_i^{\text{final}, B}).$$

Then compute the split-conformal quantile

$$\hat{q}_{\alpha, B} = \text{Quantile}_{\lceil (n_B + 1)(1 - \alpha) \rceil / n_B}(\{s_i^{(B)}\}_{i=1}^{n_B}).$$

At test time, the APS diagnostic set is

$$\Gamma_{\alpha, B}(h) = \left\{ f \in \mathcal{F} : \sum_{j: \pi_{(j)}(h) \geq \pi_f(h)} \pi_{(j)}(h) \leq \hat{q}_{\alpha, B} \right\}.$$

In implementation, every budget $B \in \{1, 2, 3, 4, \text{full}\}$ gets its own threshold $\hat{q}_{\alpha, B}$, and ProActive is frozen **before** those final thresholds are computed. This directly follows conformal classification practice and the professor’s recommendation that calibration live on a held-out split and apply to the final stopped states. [filecite turn 0 file 2](#) [cite turn 12 academia 0 turn 12 academia 1](#)

Why APS first and not RAPS: APS is the better starting point because it is simpler, already well-matched to multiclass set prediction, and is easier to debug under per-budget calibration. **RAPS** is appropriate as an ablation or fallback if APS produces many large sets; the original image-classification conformal paper introduces regularization specifically to stabilize set efficiency. [cite turn 12 academia 0 turn 16 academia 0](#)

Calibration under shift

The final plan should make a careful distinction:

- **In-distribution grouped test split:** claim nominal marginal coverage
- **Leave-model-out and held-out dataset shifts:** report **empirical coverage only**
- **Optional appendix:** add weighted or group-conditional recalibration experiments if a small labeled calibration pool from the target domain is allowed

Classical split conformal guarantees rely on exchangeability or equivalent assumptions; those assumptions are fragile under model-family or dataset shift. Therefore the paper should **not** claim formal 90% or 95% coverage on PRE-HAL, IllusionBench, or held-out models unless target-like calibration data is explicitly introduced. [\[cite\]turn12academia1\[turn18academia1\]](#)

Data, probes, labels, and features

Probe definitions and perturbation parameters

The probe family should inherit HalluPrism’s exact operational definitions wherever possible:

Probe	Implementation	Cost
blank	replace image with uniform blank canvas at same resolution	1
blur	Gaussian blur, $\sigma = 8, 15, 22$	1
crop	center crop to 35%, 50%, 65% of width and height, resize back	1
brightness	brightness factor 0.15, 0.3, 0.5	1
noise	Gaussian noise $\mu = 0, \sigma = 10, 25, 50$	1
grounding	“describe what you see, then answer” re-check prompt	1
relation	swap explicit relation term and re-ask	1
STOP	halt acquisition	0

Note, 3 different values for 4 visual probes taken in order to avoid reviewer saying, we have done hyperparameter hijacking. These settings come directly from HalluPrism’s methodology and appendix prompt catalog. Reusing them protects comparability and keeps the teacher aligned with prior evidence rather than introducing an uncontrolled new probe family. [\[filecite\]turn0file0\[turn7academia1turn19academia0\]](#)

The relation-swap action should be strictly **applicability-masked**. It is allowed only when a single reversible relation can be identified from a vetted relation lexicon: [$\{\text{\textit{\text{left}}}\rightarrow\text{\textit{\text{right}}}\}, \text{\textit{\text{above}}}\rightarrow\text{\textit{\text{below}}}\}, \text{\textit{\text{in front of}}}\rightarrow\text{\textit{\text{behind}}}\}, \text{\textit{\text{on top of}}}\rightarrow\text{\textit{\text{under}}}\}.$] Examples that contain multiple competing relation terms, negation, or non-reversible paraphrases should set `relation_applicable = 0`. This matches HalluPrism’s own restriction that relation swapping is available only on explicitly relation-bearing inputs such as VSR. [\[filecite\]turn0file0\[turn7academia1turn19academia0\]](#)

GQA relation slice

The GQA slice should be generated, not taken as raw open-ended QA. The recommended procedure is:

1. Read GQA scene-graph triples ((s,r,o))
2. Keep only triples whose relation (r) belongs to the reversible relation lexicon above
3. Generate one **positive** binary statement and one **swapped negative** statement
4. Keep only examples where the round-trip check `swap(swap(statement)) == statement` passes
5. Sample 1,000 items roughly balanced across relation types

This gives ProActive a larger relation-bearing source than VSR alone and makes the relation probe mechanically consistent across the core suite. GQA’s scene-graph-backed functional programs are exactly what makes this controlled slice feasible. [\[cite\]turn19academia0\[turn7academia1\]](#)

Teacher cache schema

Each teacher record should be a single JSONL line with the following top-level schema:

```
{
  "instance_id": "vsr_000123",
  "dataset": "VSR",
  "split": "core_train",
  "model_id": "Qwen/Qwen3-VL-8B-Instruct",
  "image_path": "data/vsr/images/...",
  "prompt_text": "...",
  "gold_answer": "false",
  "relation_applicable": true,
  "clean": {
    "raw_answer": "True",
    "norm_answer": "true",
    "correct": 0,
    "answer_logprob": -0.417,
    "answer_prob": 0.659,
    "token_entropy_mean": 0.731,
    "token_margin_mean": 0.441,
    "answer_len_tokens": 1,
    "latency_ms": 3512
  },
  "probes": {
    "blank": {...},
    "blur": {...},
    "crop": {...},
    "brightness": {...},
    "noise": {...},
    "grounding": {...},
    "relation": {...}
  },
  "teacher_signature": {"V": 0.42, "L": 0.78, "A": 0.67},
  "teacher_bits": {"visual": 0, "language": 1, "alignment": 1},
  "teacher_label6": "mixed",
  "aux_failure_family": "spatial_relation_error"
}
```

Each probe payload should contain `raw_answer`, `norm_answer`, `exact_match`, `semantic_match`, `flip`, `conf_shift`, `entropy_shift`, `margin_shift`, `applicable`, `latency_ms`, and `probe_metadata`.

The most important operational rule is: **store only aggregate features and normalized answers, not full logits or full hidden states by default**. Full logits create storage blow-up and are not needed for ProActive v1. If deeper introspection is added later, it should be isolated as an optional appendix feature family.

Exact feature extraction per model

The clean and probe feature extractor should use one unified recipe across all models:

1. Generate the answer greedily with deterministic decoding
2. Extract token-level scores from generation if the API exposes them
3. If not exposed, re-run a teacher-forced forward pass on the prompt plus generated answer and compute per-token log-probabilities from the returned logits
4. Aggregate:
 - answer probability (c)
 - mean token entropy ($\bar{H}$)
 - mean token margin ($\bar{m}$)
 - answer length (ℓ)
 - latency
 - normalized output string

For Qwen, the HF card shows standard `Qwen3VLForConditionalGeneration` loading and generation in Transformers, with explicit support for `flash_attention_2`. For Gemma, the HF card exposes multimodal loading and response parsing via the current Transformers path. For InternVL3-9B, the HF card recommends `transformers>=4.37.2`, `AutoModel`, `trust_remote_code=True`, and bf16/fp16 loading, which means score extraction should be treated as a one-week pilot target early in the project.

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Answer normalization and semantic match

Correctness and exact probe matching should be **dataset-specific but deterministic**:

Dataset	Normalization rule
POPE	map to <code>yes</code> / <code>no</code>
HallusionBench	map to <code>yes</code> / <code>no</code>
VSR	map to <code>true</code> / <code>false</code>
GQA slice	map to <code>true</code> / <code>false</code>
VizWiz-VQA	lowercase, strip punctuation/articles, collapse whitespace, map unanswerable paraphrases to <code>unanswerable</code>

For semantic probe matching, use a two-stage rule:

$$m^{\text{sem}} = \mathbf{1}[\text{exact match}] \vee \mathbf{1}[\cos(\phi(a), \phi(b)) \geq 0.82]$$

with ϕ implemented using a lightweight sentence embedding model for short English answers. This semantic fallback is relevant mainly for VizWiz free-form cases; all binary datasets should use exact normalized equality as the authoritative correctness signal.

Human audit protocol

A strong, feasible audit for ProActive is **180 examples**, not 200 or 400. The target should be:

- 30 examples targeted from each of the six teacher classes
- 3 annotators
- blind to dataset name, model name, scores, and teacher label
- shown: image, prompt, gold answer, clean answer, and a compact table of probe answers without numeric source scores

This is a **behavioral-label audit**, not a causal-source audit. It should validate whether the six-way rule-backed labels are semantically interpretable given the observed probe behavior. HalluPrism’s own audit found only modest agreement for fine-grained source judgments, which is exactly why ProActive should audit the teacher label semantics but should not base the main paper on a claim of human-ground-truth causal recovery. [fileciteopen0file00](#)

The audit acceptance thresholds should be:

Metric	Minimum acceptable
Fleiss’ (κ) six-way	0.20
At-least-two-of-three agreement	75%
Agreement of rule label with adjudicated label	55% overall
Agreement on singleton labels only	65%

If the full six-way agreement is weak but singleton-label agreement is acceptable, keep the six-way task and explicitly present “mixed” and “unclear” as ambiguity-preserving categories rather than mistakes.

Model architecture, training pipeline, and repository

Recommended architecture

The main network should remain deliberately small.

Clean token encoder

$$e_{\text{clean}} = \text{MLP}_{\text{clean}}(g(x)) \in \mathbb{R}^{64}.$$

Probe token encoder

$$e_{a_t, o_t} = \text{MLP}_{\text{probe}}([\text{Emb}(a_t), \psi(o_t), \text{Emb}(B_t)]) \in \mathbb{R}^{64}.$$

Sequential encoder

$$h_t = \text{GRU}(h_{t-1}, e_{a_t, o_t}), \quad h_t \in \mathbb{R}^{128}.$$

Heads

$$p_\theta(f \mid h_t) = \text{softmax}(\text{MLP}_6(h_t)),$$

$$r_\theta(b \mid h_t) = \sigma(\text{MLP}_3(h_t)),$$

$$\hat{s}_\theta(h_t) = \text{MLP}_{\text{VLA}}(h_t),$$

$$\hat{y}_{\text{VOI}}(h_t, a) = \text{MLP}_{\text{VOI}}([h_t, \text{Emb}(a), \text{Emb}(B_t)]).$$

Recommended hidden sizes:

Module	Size
Clean MLP	64
Probe MLP	64
GRU hidden	128
GRU layers	1
Diagnostic head hidden	64
VOI head hidden	64

This is intentionally modest. The goal in nine weeks is not to win by scale; it is to prove that active diagnostic acquisition works at all.

Training pipeline

The full pipeline should be implemented in the following order:

1. **Clean inference harness**
2. **Teacher probe generation**
3. **Teacher label construction**
4. **State sampler**
5. **Diagnostic model**
6. **Temporary validation-time APS**
7. **VOI target generation**
8. **Policy training**
9. **Final frozen per-budget APS calibration**
10. **Evaluation, transfer, ablation, freeze**

The teacher and policy should use a **sampled state dataset**, not exhaustive enumeration. With 7 atomic probes and budgets up to 4, the number of ordered prefixes becomes large enough to waste time. The state sampler should therefore generate:

- all singleton probe states
- all baseline-order prefixes
- 16 random legal trajectories per instance
- 8 policy-imitation trajectories per instance after the first policy checkpoint

That gives a tractable, diverse partial-state corpus without implementing an intractable exhaustive tree.

Repository structure

```
proactive/
  README.md
  pyproject.toml
  requirements.txt
  configs/
    data/
      hallucinationbench.yaml
      pope.yaml
      vizwiz.yaml
      vsr.yaml
      gqa_relation.yaml
      prehal.yaml
      illusionbench.yaml
    models/
      qwen3_vl_8b.yaml
      gemma4_e4b.yaml
      internvl3_9b.yaml
    experiments/
      teacher_core.yaml
      diag_train.yaml
      policy_train.yaml
      calibrate_aps.yaml
      eval_frontier.yaml
  src/proactive/
    data/
      loaders.py
      gqa_relation_builder.py
      splits.py
    prompts/
      templates.py
    probes/
      blank.py
      visual.py
      grounding.py
      relation.py
      apply.py
    models/
      qwen_adapter.py
      gemma_adapter.py
      internvl_adapter.py
      score_extraction.py
    features/
      normalization.py
      semantic_match.py
      image_quality.py
      question_type.py
    teacher/
      run_clean.py
      run_teacher.py
      build_labels.py
      sample_states.py
    networks/
      encoders.py
      diag_head.py
      voi_head.py
    conformal/
      aps.py
      raps.py
      calibration.py
    train/
      train_diag.py
      build_voi_targets.py
      train_policy.py
    eval/
      baselines.py
      frontier.py
      calibration_metrics.py
      transfer.py
      latency.py
      plots.py
    utils/
      io.py
```

```

seed.py
logging.py
manifest.py
scripts/
run_clean.py
run_teacher.py
build_labels.py
sample_states.py
train_diag.py
build_voi_targets.py
train_policy.py
calibrate_aps.py
eval_frontier.py
eval_transfer.py
run_human_audit_export.py
measure_latency.py
tests/
test_normalization.py
test_relation_swap.py
test_visual_probes.py
test_teacher_schema.py
test_state_sampler.py
test_aps.py
test_policy_masks.py

```

Major scripts and CLI contracts

Script	Purpose	Inputs	Outputs
<code>run_clean.py</code>	clean inference only	dataset manifest, model config	<code>clean_*.jsonl</code>
<code>run_teacher.py</code>	all probes for teacher cache	clean jsonl, probe config	<code>teacher_*.jsonl</code>
<code>build_labels.py</code>	six-way and bit labels	teacher jsonl	<code>labels_*.jsonl</code>
<code>sample_states.py</code>	build partial-state corpus	teacher jsonl, budgets	<code>states_*.jsonl</code>
<code>train_diag.py</code>	GRU diagnostic model	states train/val	<code>diag.ckpt</code> , <code>diag_metrics.json</code>
<code>build_voi_targets.py</code>	realized VOI targets	frozen diag model, val APS	<code>voi_*.jsonl</code>
<code>train_policy.py</code>	train ProActive policy	VOI corpus	<code>policy.ckpt</code>
<code>calibrate_aps.py</code>	final per-budget APS thresholds	frozen diag+policy, cal split	<code>aps_thresholds.json</code>
<code>eval_frontier.py</code>	main figures/tables	frozen stack, test split	<code>frontier.csv</code> , plots
<code>eval_transfer.py</code>	LOMO and held-out shifts	frozen stack, transfer split	<code>transfer.csv</code> , plots
<code>measure_latency.py</code>	GPU-time measurement	frozen stack, eval subset	<code>latency.csv</code>

Recommended CLI pattern:

```

python scripts/run_teacher.py \
  --config configs/experiments/teacher_core.yaml \
  --model qwen3_v1_8b \
  --dataset hallucinationbench \
  --split train \
  --out outputs/teacher/qwen3_v1_8b/hallucinationbench_train.jsonl

```

Every script should support:

- `--seed`
- `--resume`
- `--limit`
- `--device`
- `--output_dir`
- `--manifest_path`

Checkpoint naming should follow:

[\texttt{(stage)_(modelset)_(dataset)_(seed)_(date).ckpt}]

Evaluation matrix, baselines, ablations, and compute

Baselines and what they answer

Baseline	Exact implementation	Reviewer objection addressed
Scalar confidence	clean pass only, threshold/ranking from (c)	“Standard uncertainty may already be enough.”
Clean-only learned	same GRU heads but no probes, only clean token	“Probing may be unnecessary.”
Clean-only conformal	APS on clean-only diagnostic model	“Sets may help without active acquisition.”
Random acquisition	random legal action under same budget	“Gains might come from probing at all, not policy learning.”
Blank-first	fixed order: blank then stop/continue	“A simple manually chosen order may be sufficient.”
Visual-first	blur→crop→brightness→noise	“Visual stress alone may explain most gains.”
Grounding-first	grounding then others	“One universal grounded re-check may suffice.”
Relation-first	relation first when applicable	“Relation-heavy tasks may not need adaptivity.”
Uncertainty-greedy	next action = max predicted entropy drop	“Heuristic acquisition may match learned VOI.”
Dataset-specific fixed	hand-tuned order per dataset	“Your policy may only be learning dataset identity.”
Full teacher	all probes, highest cost	Upper-bound reference
Oracle next probe	best next action from realized teacher effect	“How far are you from the budgeted optimum?”
Oracle best subset	best subset up to budget (B) when feasible	Ceiling for subset selection
One-pass distilled	train direct clean predictor to imitate full teacher	“This is really just distillation, not acquisition.”

The professor’s document explicitly calls for scalar, clean-only, fixed-order, random, full-probe, oracle, and conformal baselines, and those are exactly the baselines that matter for reviewer-resistant evaluation. □filecite□turn0file2□

Metrics

The evaluation should report five metric families.

Family	Metric	Primary
Diagnostic class quality	Macro-F1, balanced accuracy, AUROC	Secondary
Diagnostic sets	Coverage @90, Coverage @95, avg set size, singleton rate, empty-set rate	Primary
Cost	avg additional forward passes, distribution of probe counts	Primary
Latency	normalized GPU-seconds relative to clean pass	Secondary
Policy behavior	action frequency, stop frequency, source-conditioned action mix	Secondary
Transfer	LOMO empirical coverage and set size, held-out shift coverage/set size	Primary
Correction utility	accuracy, fix rate, break rate, regret	Appendix / secondary

Define

$$\bar{S}_{\alpha,B} = \frac{1}{N} \sum_{i=1}^N |\Gamma_{\alpha,B}(h_{T,i})|, \quad \text{Coverage}_{e_{\alpha,B}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[f_i \in \Gamma_{\alpha,B}(h_{T,i})].$$

The main frontier should plot **average additional forward passes** on the x-axis and **average calibrated set size at 90% coverage** on the y-axis, with a secondary frontier plotting **Macro-F1** versus mean additional passes.

Define the frontier integrals by trapezoidal approximation over method points sorted by average cost (C_j):

$$\text{AUC}_{\text{diag}} = \sum_j \frac{D_j + D_{j+1}}{2} (C_{j+1} - C_j),$$

where $D_j = \text{MacroF1}_j$, and

$$\text{AUC}_{\text{set}} = \sum_j \frac{E_j + E_{j+1}}{2} (C_{j+1} - C_j), \quad E_j = 1 - \frac{S_{\alpha,j}}{6}.$$

This gives one scalar summary for accuracy–cost and one for set-efficiency–cost.

Ablations

Ablation	Purpose
No budget embedding	test whether budget conditioning matters
No STOP action	test whether stopping policy is necessary
No VOI loss	test whether acquisition learning matters
Loss-only VOI	compare against set-shrinkage objective
No answer-flip features	test probe discrete instability information
No confidence-shift features	test soft evidence information
No semantic match	test whether lightweight semantics helps
No relation probe	test relation-specific value
Bundled visual probes	compare atomic vs bundled probing
APS vs RAPS	efficiency under coverage
Global vs per-budget calibration	calibration design check
Pass-count vs latency cost	cost robustness
Independent heads	test whether entanglement-preserving representation matters
No signature regression	auxiliary teacher reconstruction check

Compute and storage estimates

With the recommended sample caps, the full teacher generation workload is approximately:

- **Core per model:** 59,377 passes
- **Held-out per model:** 9,000 passes
- **Total per model:** 68,377 passes
- **Total across 3 models:** 205,131 passes

These totals assume 7 passes for non-relation examples and 8 for relation-bearing examples, consistent with HalluPrism’s protocol.

Using a conservative end-to-end **4–8 seconds per pass** for 7–9B-class models on an A6000, the teacher cache phase alone is roughly:

Mean pass time	Total GPU-hours
4 s	228 h
6 s	342 h
8 s	456 h

That is feasible within nine weeks if the project uses one or more A6000s continuously for teacher generation during Weeks 3–5. If only one GPU is reliably available, the plan should keep the sample caps above and avoid late-scale expansions. The teacher phase is almost certainly more time-consuming than GRU or conformal training. This is exactly why the professor’s document recommends early pilot generation and hard compute-aware gating.

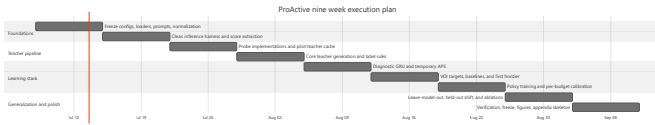
A realistic storage estimate for compressed JSONL probe caches is:

Artifact	Estimated size
Teacher JSONL caches	2–6 GB
State corpus	5–15 GB
Checkpoints and plots	5–10 GB
Logs and manifests	<2 GB
Total recommended free space	30–50 GB

This estimate assumes you **do not** store full logits or permanently store all perturbed image files. Perturbed images should be generated deterministically on demand from the original image plus a stored probe spec.

Nine week schedule

Timeline overview



Week by week execution table

Week	Objective	Prerequisites	Exact tasks	Expected outputs	Tests	GPU workload	Completion gate
Week 1	Freeze implementation skeleton	Frozen decisions	Finalize dataset manifests, prompt templates, normalization rules, model adapters, and deterministic decoding settings	<code>configs/*</code> , <code>loaders.py</code> , <code>templates.py</code> , <code>normalization.py</code> , reproducibility manifest	unit tests for normalization; prompt round-trip tests	0–5 h	All datasets parse; all model configs compile
Week 2	Clean inference harness	Week 1	Implement clean generation for all 3 models; expose answer prob, entropy, margin, length, latency; run 50-example smoke tests per dataset/model	<code>clean_*.jsonl</code> , <code>run_clean.py</code> , <code>score_extraction.py</code>	string normalization audit; token-score sanity checks; deterministic rerun test	15–30 h	All 3 models produce stable clean outputs on all core datasets
Week 3	Teacher probes and pilot	Week 2	Implement blank, 4 visual probes, grounding, relation; build teacher schema; generate 100-example pilot per core dataset per model	<code>teacher_pilot/*.jsonl</code> , <code>probe</code> modules, probe timing report	perturbation visual sanity checks; relation round-trip tests; schema validation	25–45 h	Pilot shows nontrivial variation in flips/conf shifts and no catastrophic saturation
Week 4	Scale core teacher and construct labels	Week 3	Run full core teacher generation; compute HalluPrism (V,L,A); build bit labels and six-way labels; export 180-example audit packet	<code>teacher_core/*.jsonl</code> , <code>labels_core.jsonl</code> , <code>audit_packet.csv</code>	label distribution checks; per-class minimum counts; pilot audit of 20 examples	80–140 h	Teacher core cache complete for all core datasets on at least 2 models; label class histogram usable
Week 5	Diagnostic GRU and temporary APS	Week 4	Sample partial states; train clean-only and GRU diagnostic models; fit temporary validation APS	<code>states_train.jsonl</code> , <code>diag.ckpt</code> , <code>clean_only.ckpt</code> , <code>aps_val_tmp.json</code>	APS coverage on val; confusion matrix inspection; no-leak split audit	15–30 h	GRU beats clean-only on partial-state diagnosis or at least matches with smaller

Week	Objective	Prerequisites	Exact tasks	Expected outputs	Tests	GPU workload	Completion gate
			for VOI target generation; run first set-size/coverage checks				partial uncertainty
Week 6	VOI targets, baselines, first frontier	Week 5	Build realized VOI targets from frozen diagnostic model; implement scalar, clean-only, random, fixed-order, full-teacher, uncertainty-greedy baselines; plot first frontier	voi_targets.jsonl , baseline_results.csv , frontier_v0.png	action-mask tests; VOI sanity tests; matched-budget validation	20–40 h	At least one probe-aware baseline beats scalar/clean-only at matched cost
Week 7	Train ProActive policy and final APS	Week 6	Train budget-conditioned policy with STOP; tune (λ_B); freeze stack; compute final per-budget APS thresholds on calibration split; generate main 90/95 coverage tables	policy.ckpt , aps_thresholds.json , frontier_main.csv	per-budget coverage; stop-rate sanity; action frequency by dataset	20–35 h	ProActive beats random and at least two fixed orders on average set-size frontier
Week 8	Generalization and ablations	Week 7	Run leave-one-model-out; run PRE-HAL and IllusionBench; execute highest-value ablations; measure latency	lomo.csv , shift.csv , ablations.csv , latency.csv	empirical coverage under shift; within-dataset policy sanity	40–80 h	LOMO remains above clean-only/random; held-out shift shows at least one meaningful gain
Week 9	Verification and freeze	Week 8	Final plots, tables, appendix skeleton, risk memo, experiment freeze; rerun main results with fixed seeds; package manifests	figures/* , tables/* , repro_manifest.json , freeze_checklist.md	full reproducibility rerun; checksum match; no leakage audit	10–25 h	Paper-ready figure set and frozen result table

Per-week expected artifacts

The project should explicitly produce the following by week:

Week	Code artifacts	Data artifacts	Model artifacts	Plots
1	loaders, prompts, normalization	manifests	none	dataset summary
2	model adapters, score extraction	clean smoke caches	none	latency smoke plot
3	probe modules	teacher pilot cache	none	probe diagnostic histograms
4	label builder, audit exporter	core teacher cache, label jsonl	none	class distribution, source-bit plots
5	GRU diagnostic trainer, APS temp	state corpus	<code>diag.ckpt</code> , <code>clean_only.ckpt</code>	first set-size/coverage plot
6	VOI target builder, baselines	VOI corpus	none	baseline frontier v0
7	policy trainer, calibration scripts	cal outputs	<code>policy.ckpt</code> , APS thresholds	main frontier, action heatmaps
8	transfer and ablation runners	transfer csvs	ablation checkpoints	LOMO plot, shift plot
9	report packer, freeze scripts	final tables	frozen named stack	camera-ready figure set

Risks, open questions, and experiment freeze checklist

Risk register

Risk	Detection	Threshold for concern	Mitigation	Fallback claim	Fatal
Clean-only nearly matches full teacher	compare set size at 90% coverage	<10% gap	revise label rules or probe semantics	“Teacher distillation is enough”	Yes
Fixed order matches ProActive	frontier gap vs fixed orders	ProActive wins <2 budgets	analyze source slices, strengthen VOI target	“Source-conditioned schedules help”	Yes
InternVL score extraction unstable	missing/invalid token scores	> 5% failures	teacher-forced scoring pass	keep InternVL for transfer only	No
Relation probe too sparse	applicability rate	<15% in relation datasets	enlarge GQA slice	rely more on grounding	No
APS sets too large	avg size at 90%	>3.2 on six-way task	try RAPS ablation	report coverage with larger sets	No
Severe undercoverage under shift	empirical coverage on held-out tests	<80% at 90% target	report as stress only; no guarantee claim	“Good stress behavior, not formal coverage”	No
Dataset leakage dominates	dataset-specific fixed baseline close to ProActive	gap <2%	remove dataset-id-like features, rebalance, report within-dataset	narrow claim to within-distribution	Yes
Human audit disagreement high	κ and agreement	$\kappa < 0.15$	merge rare classes or relax thresholds	“Behavioral target remains heuristic”	No
Teacher compute overruns	actual GPUh vs estimate	>25% over plan	cap VizWiz and held-out size first	smaller but still valid evaluation	No
ProActive beats random but not fixed	frontier table	only 1 fixed baseline beaten	improve state features, stopping target	source-dependent schedule paper	Borderline
Held-out source shift weak	PRE-HAL / IllusionBench gains	no gains anywhere	emphasize LOMO and in-distribution frontier	shift as negative finding	No
Calibration split contaminated	leaked instances across splits	any overlap	rebuild split with grouped hashing	rerun final metrics only	Yes

Explicit open questions that should remain open until pilot completion

These are the only items that should remain explicitly open in the project document:

Open question	Why it remains open	Planned resolution point
Exact pinned checkpoint revisions for all 3 models	model cards can update; exact revision was intentionally not frozen now	end of Week 1
InternVL token-score extraction path	custom <code><trust_remote_code></code> can expose logits differently	Week 2
Final semantic-match threshold for VizWiz	short-answer similarity threshold should be pilot-tuned	Week 3
Exact GQA relation slice mapping table	generated statements need manual validation	Week 3
Whether APS needs RAPS backup	depends on observed set sizes in Week 5	Week 5

Everything else in this report should be treated as the default implementation specification.

Experiment-freeze checklist

The project should freeze only when all items below are satisfied:

- grouped train/val/cal/test split checksum saved
- all three models pinned to exact revisions
- teacher cache manifest complete and reproducible
- six-way label rules and thresholds frozen
- clean-only, random, fixed-order, uncertainty-greedy, full-teacher, and oracle baselines completed
- final ProActive checkpoint selected on validation only
- APS thresholds computed on calibration split only
- 90% and 95% coverage tables regenerated from scratch
- leave-one-model-out table regenerated from scratch
- shift tables regenerated from scratch
- latency measurement rerun with fixed hardware settings
- leakage audit passed
- human-audit summary attached
- risk memo written
- no missing seeds, manifests, or output schemas

Final go and no-go criteria

The professor's go/no-go logic should remain the final decision rule:

1. ProActive must beat **random acquisition** at matched budget.
2. It must beat **at least two fixed probe orders** on the average frontier.
3. **Clean-only** must not match **full probing**.
4. Diagnostic sets must **shrink meaningfully after adaptive probing** while maintaining target coverage.
5. **Leave-one-model-out** ProActive must remain above scalar confidence and clean-only baselines.
6. Full probing must remain the strongest **upper-bound reference** for mixed or hard cases.
7. The paper must be explainable without assuming the reader already knows HalluPrism. □filecite□turn0file2□

If the first four fail by the end of **Week 7**, the ICLR-aimed "active diagnostic sets" claim should be downgraded. If leave-one-model-out fails by the end of **Week 8**, the paper should be reframed around within-distribution active diagnosis rather than broad model-general acquisition. That is the fairest, most reviewer-resistant interpretation of the planning document and the strongest way to protect the project's remaining time. □filecite□turn0file2□